CMPG311 Group Project

Royal Fields Database System Proposal

Phase 1

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1. MEMBERS OF THE GROUP

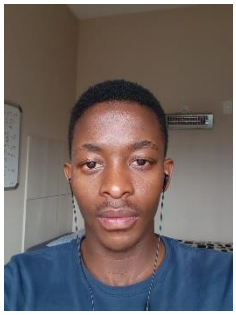
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2. COMPANY SITUATION ANALYSIS

Company Scenario:

Royal Fields is a community-supported agriculture (CSA) initiative operating in select areas of Limpopo Province, South Africa. It aims to connect local small-scale farmers with consumers in the region, providing access to fresh, culturally relevant produce. Limpopo, with its diverse agricultural potential, offers a unique opportunity to promote sustainable farming practices and support local economies

2.1 COMPANY OBJECTIVES (MISSION OF THE ENVIRONMENT)

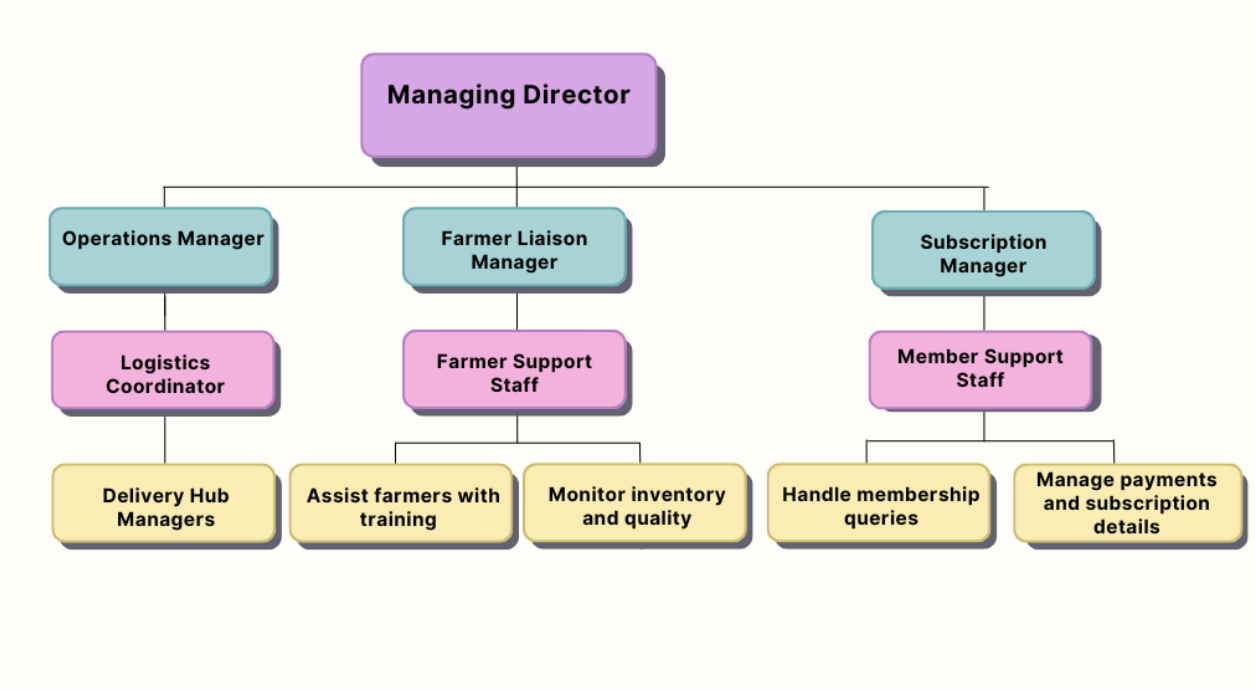
- **Regional Accessibility:** Expand access to fresh, locally grown produce within Limpopo Province.
- **Support for Small-Scale Farmers:** Empower and provide market access to emerging and small-scale farmers in the region.
- **Culturally Relevant Produce:** Ensure the availability of produce that aligns with the dietary preferences and cultural traditions of the local communities.
- **Efficient Logistics:** Manage the distribution of produce across the geographically dispersed areas of Limpopo.
- **Community Engagement:** Foster a sense of community and promote awareness of sustainable agriculture.

2.2 COMPANY OPERATIONS (GENERAL OPERATING ENVIRONMENT)

- Members subscribe to receive weekly or bi-weekly boxes of seasonal produce.
- Farmers supply produce to Royal Fields, adhering to quality and cultural relevance standards.
- Royal Fields staff coordinate farmer relationships, manage subscriptions, plan delivery logistics and handle financial transactions.
- Produce boxes contain a variety of seasonal produce, including traditional African vegetables and fruits.
- Strategic delivery hubs are established across Limpopo to facilitate efficient distribution.
- Flexible payment options are implemented, considering the diverse economic circumstances of the region.

2.3 COMPANY ORGANIZATIONAL STRUCTURE

Royal Fields is managed by a core team responsible for the overall operations. This team handles the day-to-day running of the initiative, including coordinating with farmers, managing subscriptions, and organizing deliveries. Royal Fields works with independent farmers who supply the produce. These farmers are not employees of Royal Fields but rather partners who contribute to the initiative. The initiative is built on a member-based consumer model, where individuals subscribe to receive produce boxes. These members are the end consumers of the produce supplied by the farmers.



2.3.1 Managing Director

The Managing Director holds the highest position within Royal Fields. This individual is responsible for the overall strategic direction and management of the organization. All other departments report directly or indirectly to the Managing Director, and significant decisions require their approval.

2.3.2 Operations Manager

The Operations Manager oversees the day-to-day operational activities of Royal Fields. This includes the management of logistics, delivery, and overall efficiency of the organization's processes.

- **Logistics Coordinator:** The Logistics Coordinator is responsible for planning and managing the delivery routes and scheduling. They ensure the smooth and timely delivery of produce boxes to members.
- **Delivery Hub Managers:** These managers are responsible for overseeing the operations of individual delivery hubs. They manage the sorting, packing, and dispatch of produce boxes from their respective hubs.

2.3.3 Farmer Liaison Manager

The Farmer Liaison Manager is the primary point of contact between Royal Fields and the local farmers. This role is crucial for maintaining strong relationships with farmers and ensuring a consistent supply of quality produce.

- **Farmer Support Staff:** This team provides support to farmers, including training on best practices and assistance with any issues they may encounter. They also monitor the inventory and quality of produce coming from the farms.
- **Assist farmers with training:** This team provides training to the farmers on best farming practices.
- **Monitor inventory and quality:** This team is responsible for quality control.

2.3.4 Subscription Manager

The Subscription Manager is responsible for managing all aspects of member subscriptions, including handling inquiries, managing subscription details and processing payments.

- **Member Support Staff:** This team handles membership queries and provides customer service to members. They also manage payment processing and subscription details.
- **Handle membership queries:** This team is responsible for answering any questions that members may have.
- **Manage payments and subscription details:** This team is responsible for the financial side of the subscriptions.

This organizational structure supports the core objectives of Royal Fields by emphasizing strong relationships with farmers, efficient logistics, and excellent customer service.

3. PROBLEMS AND CONSTRAINTS

3.1 PROBLEMS:

- Geographical challenges: Limpopo's vast area and varying infrastructure pose logistical challenges for delivery.
- Farmer capacity: Many small-scale farmers may have limited capacity for consistent supply and quality control.
- Market access: Lack of reliable market access for small-scale farmers.
- Technology adoption: Varying levels of technology adoption among farmers and consumers.
- Financial constraints: Limited funding and resources for operational expansion.
- Cultural Sensitivity: Ensuring produce selection and distribution align with local cultural preferences.
- Transportation: Limpopo is a large area, so transportation costs could be high.

3.2 CONSTRAINTS:

- **Budgetary constraints:** The project must be completed using existing resources, avoiding expensive software purchases.
- **Time constraints:** The project must be completed within the semester timeframe.
- **Personnel constraints:** The database needs to be able to be operated by people with basic computer skills.

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4. DATABASE SYSTEM SPECIFICATION

4.1 DEFINE OBJECTIVES TO SOLVE PROBLEMS IDENTIFIED:

- Develop a database to automate subscription management, inventory tracking, and farmer communication.
- Create a centralized database to store and manage all relevant data.
- Implement database features to facilitate efficient communication and reporting.
- Design a database to accurately track payments and generate financial reports.
- Design a database that can be expanded in the future.
- Design a database that can optimize delivery routes.

4.2 INFORMATION THAT COMPANY REQUIRES FROM DATABASE:

- Member information (contact details, subscription details, payment history).
- Farmer information (contact details, produce inventory, payment details).
- Produce information (types, quantities, prices, cultural relevance).
- Subscription details (start/end dates, delivery schedules).
- Delivery information (routes, statuses, hubs, driver information, delivery times).
- Payment records (member payments, farmer payouts).
- Inventory records (available quantity, harvest date, expiry date).
- Reports relating to income, expenses and farmer payments.
- Event information.

- The database will manage member subscriptions, farmer information, produce inventory, delivery logistics and financial transactions.
- The database will support basic reporting functions and community engagement features.
- The project will be limited to the creation of the database and will not include the creation of a user interface.

4.3 BOUNDARIES:

- The project will not include advanced features such as real-time vehicle tracking or online farmer training modules.
- The project will focus on the core database design and implementation.
- The project will be contained to the functions of the database and will not include any website design.
- The project will be contained to the functions of the database and will not include any mobile app design.
- The project will be limited by the time constraints of the semester.
- The project will be limited by the budget constraints of the group.

5. CONCLUSION

The Royal Fields initiative plays a vital role in connecting small-scale farmers in Limpopo with consumers, fostering sustainable agriculture and community engagement. However, logistical challenges, financial constraints, and the need for efficient management systems pose significant obstacles. To address these challenges, the development of a robust database system is essential.

The proposed database will streamline subscription management, inventory tracking, farm communication and financial transactions. By implementing a centralized system, Royal Fields can improve efficiency, optimize delivery logistics, and enhance market access for farmers. While the project is constrained by budget, time, and personnel limitations, its successful implementation will create a foundation for future scalability and operational improvements. This database-driven approach will ultimately contribute to the sustainability and growth of Royal Fields, ensuring fresh, culturally relevant produce reaches communities across Limpopo.